



BUSINESS
SCHOOL

SKEMA

Derivatives and Fixed Income PGE M1 Fall 2026

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Session 9(12): Option Strategies



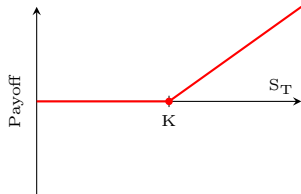
About me

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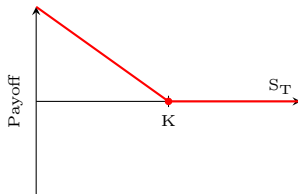
Payoff Diagrams: Calls and Puts



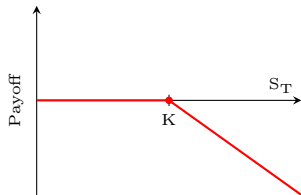
Long Call



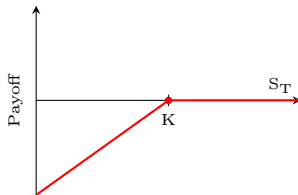
Long Put



Short Call



Short Put



Payoff Tables



	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Put	$-P$	$K - S_T$	0
Short Put	$+P$	$S_T - K$	0
Long Call	$-C$	0	$S_T - K$
Short Call	$+C$	0	$K - S_T$

Note, for a long call position, a payoff of $\max\{S_T - K, 0\}$ is equivalent to a payoff of $S_T - K$ when $S_T > K$ and 0 otherwise, as the option is not exercised.

Payoff Tables



	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Stock	$-S_0$	S_T	S_T
Short Stock	$+S_0$	$-S_T$	$-S_T$
Borrow (sell bond)	$+S_0$	$-S_0(1+r)^T$	$-S_0(1+r)^T$
Lend (buy bond)	$-S_0$	$S_0(1+r)^T$	$S_0(1+r)^T$
Borrow (sell bond)	$+\frac{K}{(1+r)^T}$	$-K$	$-K$
Lend (buy bond)	$-\frac{K}{(1+r)^T}$	K	K

Technically, you can borrow and lend any amount today but S_0 and $\frac{K}{1+r}$ are the ones relevant.



Some Terminology

An option is **at-the-money (ATM)** if the stock price today is equal to the strike price, $S_0 = K$.

An option is **in-the-money (ITM)** when the payoff of immediate exercise is positive.

An option is **out-of-the-money (OTM)** when the payoff of immediate exercise is negative.



A **straddle** is an options trading strategy that is a combination of a long call and a long put at the same strike price K and expiration date T . Assume you enter a straddle on Apple stock at $K = 350$ with expiry in 1 year. Assume $r = 5\%$, and $c = 14.29$, $p = 23.81$. For simplicity, assume Apple does not pay dividends.

- a. Find the combined payoff of this strategy using tables
- b. Plot the payoff diagram
- c. Plot the profit diagram



- a. Find the combined payoff of this strategy using tables.

	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Call			
Long Put			
Total Payoff			

For payoff at date T, you want $\#_K + 1$ columns, where $\#_K$ is the number of strikes

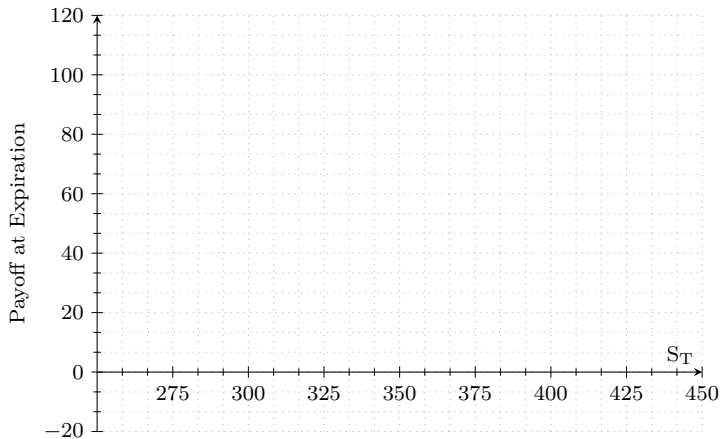


a. Find the combined payoff of this strategy using tables.

	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Call	-14.29	0	$S_T - 350$
Long Put	-23.81	$350 - S_T$	0
Total Payoff	-38.10	$350 - S_T$	$S_T - 350$

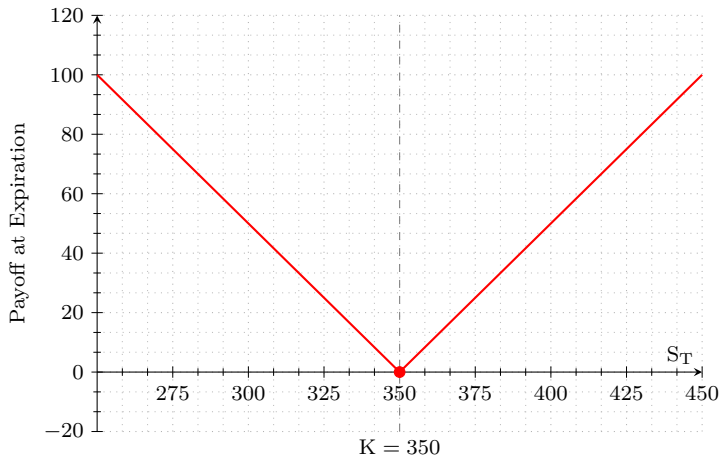


b. Plot the payoff diagram.





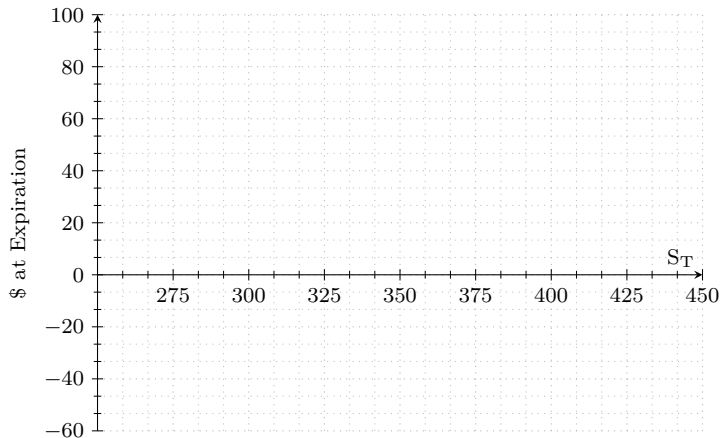
b. Plot the payoff diagram.



Straddle



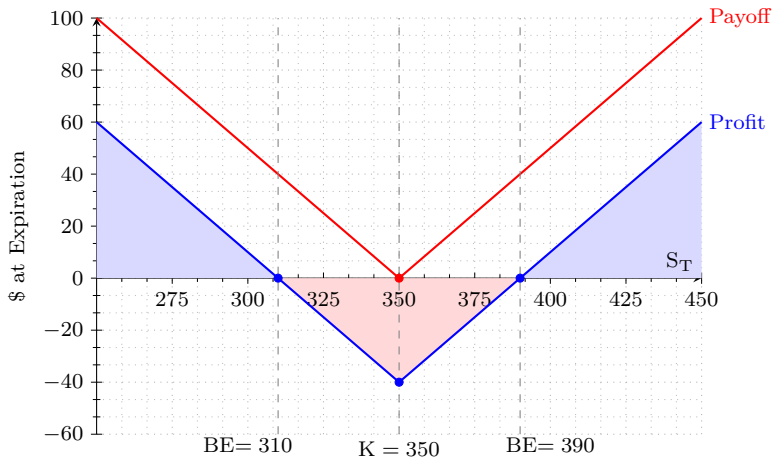
c. Plot the profit diagram. Assume you paid a combined premium of \$38.10 for the call and put ($38.10 \times 1.05 = 40$).



Straddle



c. Plot the profit diagram. Assume you paid a combined premium of \$38.10 for the call and put ($38.10 \times 1.05 = 40$).



Who would trade a straddle?





Bull Call Spread

A **bull call spread** is a combination of a long position in a call option with a lower strike price K_1 and a short position in a call with a higher strike K_2 , such that $K_1 < K_2$. Assume $r = 5\%$, $T = 1$ year for both options, Apple does not pay dividends. Let $K_1 = 320$ and $K_2 = 380$, and the prices of calls are $c_1 = 52.3809$, $c_2 = 25.7143$.

- a. Find the combined payoff of this strategy using tables
- b. Plot the payoff diagram
- c. Plot the profit diagram

Bull Call Spread



a. Find the combined payoff of this strategy using tables.

	Today	Payoff at T		
		$S_T < K_1$	$K_1 < S_T < K_2$	$S_T > K_2$
Long Call (K_1)				
Short Call (K_2)				
Total Payoff				

For payoff at date T, you want $\#_K + 1$ columns, where $\#_K$ is the number of strikes. Here, we need 3 columns.

Bull Call Spread



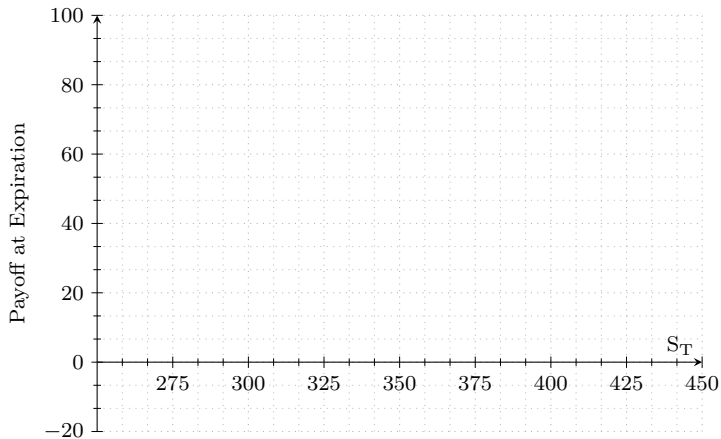
a. Find the combined payoff of this strategy using tables.

Today		Payoff at T		
		$S_T < K_1$	$K_1 < S_T < K_2$	$S_T > K_2$
Long Call (K_1)	-52.3809	0	$S_T - 320$	$S_T - 320$
Short Call (K_2)	+25.7143	0	0	$380 - S_T$
Total Payoff	-26.6666	0	$S_T - 320$	60

Bull Call Spread



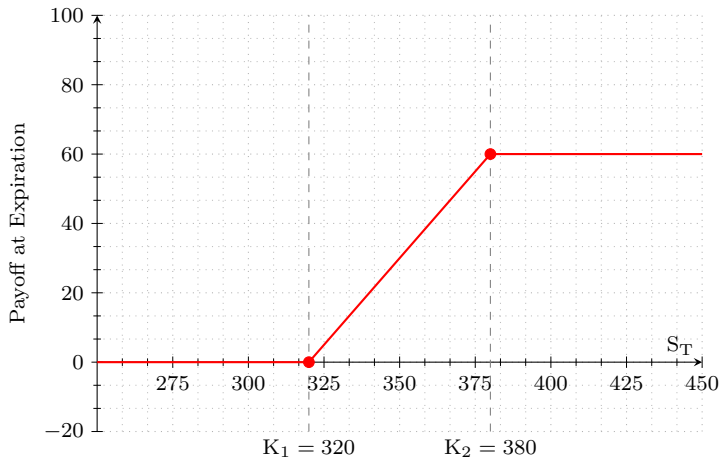
b. Plot the payoff diagram.



Bull Call Spread



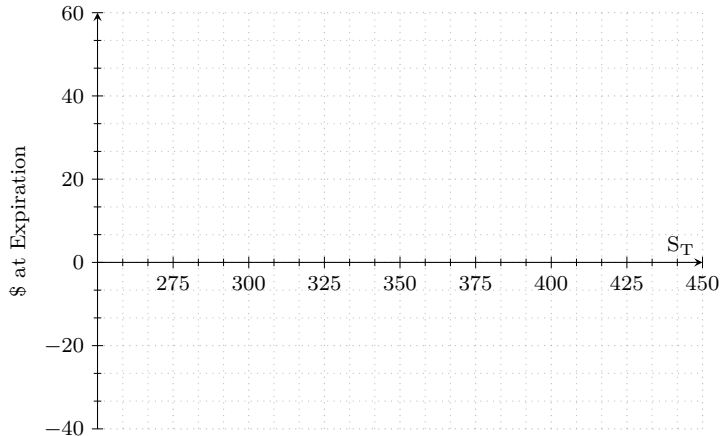
b. Plot the payoff diagram.



Bull Call Spread



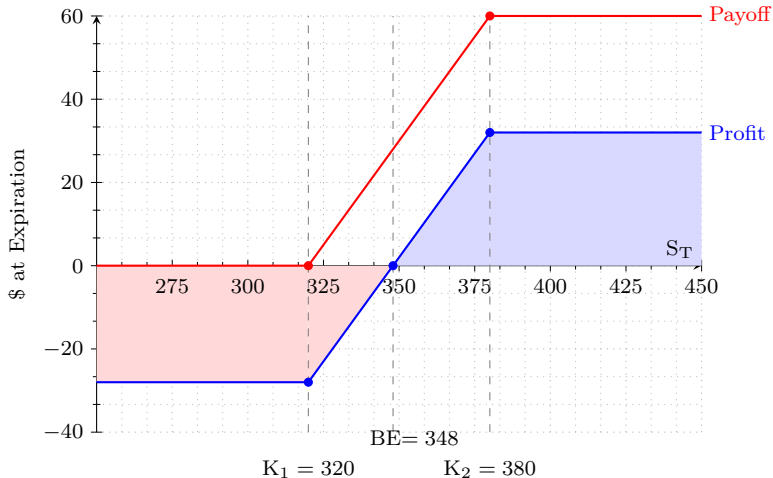
c. Plot the profit diagram. Assume you paid a net premium of \$26.6666 ($26.6666 \times 1.05 = 28$).



Bull Call Spread



c. Plot the profit diagram. Assume you paid a net premium of \$26.6666 ($26.6666 \times 1.05 = 28$).



Who would trade a call spread?



Excel



On K2, find the file “DFI_AAPL_Options_Grid.xlsx”.

Covered Call



Combine a strategy of long stock plus a short call written at $K = 360$.
Compare the max profit/loss when $K = 340$ and $K = 380$ instead, and
to the max profit/loss of holding only the underlying.



Protective Put

Combine a strategy of long stock plus a long put written at $K = 320$.

Compare the max profit/loss and option premiums when $K = 300$ and $K = 340$ instead.



Bull Call Spread

Combine a long call at $K_1 = 320$ and a short call at $K_2 = 360$.

Compare the net premium to a standalone long call at $K = 320$.

Bull Call Ladder



Combine a long call at $K_1 = 320$, a short call at $K_2 = 340$, and a short call at $K_3 = 360$.

What is the trade-off of the bull call ladder compared to the bull call spread?



Bear Put Spread

Combine a long put at $K_1 = 360$ and a short put at $K_2 = 320$.

Compare the net premium to a standalone long put at $K = 360$.



Long Straddle

Combine a long call and long put at $K = 340$.

Almost always, both legs of the trade are initiated ATM.

On the **Inputs** tab,



Long Strangle

Combine a long call at $K_1 = 360$ and a long put at $K_2 = 320$.

(Generally, the long strangle is a combination of an OTM call and an OTM put.)

Compare the net premium to that of a long straddle. Comment on the relative price of OTM vs ATM options.

Short Straddle



Combine a short call and short put at $K = 340$. Compare the max profit/loss to the Long Straddle at $K = 340$. What is the market view of someone trading a short straddle?

Iron Condor



Combine a long put at $K_1 = 300$, a short put at $K_2 = 320$, a short call at $K_3 = 360$, and a long call at $K_4 = 380$. Compare the net premiums and max profit/loss to that of a short strangle (short put at $K_2 = 320$ and short call at $K_3 = 360$).

The short put (K_2) and short call (K_3) are both OTM options, sold to collect the option premia; the long put (K_1) and long call (K_4) are further-OTM options bought cheaply to cap the downside risk.

Iron Condor



Combine a long put at $K_1 = 300$, a short put at $K_2 = 320$, a short call at $K_3 = 360$, and a long call at $K_4 = 380$. Compare the net premiums and max profit/loss to that of a short strangle (short put at $K_2 = 320$ and short call at $K_1 = 360$).

(There is another trading strategy called the long call condor, which is a long call at a lower strike, a short call at a middle strike, another short call at the next middle strike, and a long call at a higher strike. All strikes are evenly apart. In contrast to the iron condor, the call condor pays a premium today, though the max profit/loss is identical to the iron condor.)

Butterfly



Combine a long call at $K_1 = 320$, two short calls at $K_2 = 340$, and one long call at $K = 360$. What view does this trading strategy express, and how does it differ from the condor?